# PORTFOLIO

**SC1153: Introduction to Biotechnology**




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## Abstract

The portfolio contains four assignments from SC1153: Introduction to Biotechnology. They included a group presentation on white (industrial) biotechnology, a documentary-style video on spider silk as a biomaterial, a career trajectory presentation on Frances Arnold, and a debate presenting the cons of de-extinction. The assignments explored collaboratively core ideas in industrial biotechnology, biomimicry and conservation science, whilst developing skills in scientific communication, critical assessment, teamwork and ethical reasoning.

## Acknowledgements

I want to take this opportunity to thank all those who helped me throughout this module. I am thankful to my lecturer Mr. Shanitha for the knowledge, comments and guidance he gave me during the semester. I would like to thank my group mates R. P. D. S. K. Chandrathilaka, L. D. Selitha, and Sewwandi H. G. I. for their cooperation, dedication and hard work during our presentations, video project and debates. I would also like to thank my classmates for their ideas and encouragement in our discussions. Finally, I would like to thank my family and friends for their constant support and encouragement throughout this process.

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## Objectives

- Assignment One: To collaboratively research, analyze and present on a specific area of biotechnology to improve scientific communication and teamwork.

- Assignment Two: Create a digital educational video about a biotechnology product to translate complex biotechnology ideas into an accessible and understandable format.

- Assignment Three: Investigating professional career paths in biotechnology: analyzing role models and developing a personal career trajectory.

- Assignment Four: Participate in a formal debate to assess social acceptance, bioethical dilemmas, and regulatory structures.

## Assignment One: Group Presentation

### 1. Assignment Description

For this assignment, our group was asked to look into a part of biotechnology and give a team presentation to the class. We picked White Biotechnology. The use of living things and special helpers called enzymes to make chemicals, materials and power instead of using old methods that rely on oil. The presentation talked about eight areas: how enzymes help in making things how tiny creatures make products how to make plastic from plants how to make fuel from plants how to make clothes in a better way how to use enzymes in making food how to clean up pollution and how to use better chemicals. Every person in our group. Shared a different part and the presentation ended with how white biotechnology affects the environment the money we spend and the way people live. It shows how we can create a more lasting world.

### 2. Full Group Presentation


1. ENZYMES CATALYSIS
2. MICROBIAL FERMENTATION


3. BIO - PLASTIC
4. BIO FUELS AND ENERGY


5. SUSTAINABLE TEXTILE PROCESSES
6. INDUSTRIAL ENZYMES IN FOOD PRODUCTION


WHITE
BIOTECHNOLOGY


Nature's
Factory


Did You Know?


WHAT IS WHITE
BIOTECHNOLOGY ?


THE NEED FOR A GREEN SHIFT:


THE 8 PILLARS ROADMAP


7. BIOREMEDIATION AND
WASTE TREATMENT


8. GREEN CHEMISTRY ALTERNATIVES





### 3. Your Reflection

Working on this presentation taught me how much coordination goes into a group project of this scale. Dividing the eight pillars of biotechnology among group members meant each of us had to research our section thoroughly enough to explain it clearly while trusting each teammate to do the same for their part. This presenter found it challenging to translate processes Like enzyme catalysis and bioremediation. Into language a general audience could follow. This challenge pushed this presenter to think carefully about how this presenter communicates science. Presenting also helped this presenter build confidence in speaking. Reviewing the group's slides afterward showed this presenter the value of using visuals and relatable everyday examples such as yoghurt or jeans to keep an audience engaged with a technical topic. Overall, this assignment strengthened this presenter's communication. It also reinforced that teamwork, when structured clearly can turn a complex subject into something coherent and compelling.

### 4. Photos



# Assignment Two: Video Creation

## 1. Assignment Description

For this assignment, our group made a video that looked at a certain biotechnology product. It started with the question: "What if the strongest material from nature could be made in a lab?" We picked spider silk as the topic. One of the most flexible materials found in nature. It's also a topic that more and more scientists are trying to copy in a lab. The video had group members talking in places and showed pictures that showed how the tiny parts of silk work and how scientists change genes to make it.

## 3. Written Summary

Spider silk is known for being incredibly strong, flexible and light. These qualities come from its protein structure. The proteins are made up of repeated sequences of amino acids. These sequences form linked molecular chains. Because it’s not practical to farm spiders on a scale the video explained how scientists use biotechnology. They take the genes that code for spider silk proteins. Insert them into other organisms. Bacteria are often used for this. The process uses DNA techniques. Once inside the bacteria the genes are. Read. The bacteria then produce the silk proteins. These proteins can be. Turned into fibers. The video showed how lab-made spider silk can be used in various ways. It can be used to make protective gear and body armor. It can also be used to create high-performance textiles that're sustainable. This kind of silk could be a better choice than synthetic fibers, like nylon or Kevlar. It offers an eco-friendly and scalable solution. Its uses could affect areas such as medicine, defense and fashion.

## 4. Reflection

Making this video pushed me to think about how to communicate biotechnology. Because we could not rely on slides we had to plan a script storyboard visuals and film ourselves narrating on camera. Writing the explanation of DNA technology so that it felt natural to speak aloud, rather than reading like a textbook, was harder than I expected. It taught me a lot about pacing and simplifying while keeping accuracy. Coordinating the shoot with the rest of the group deciding who would narrate each section, filming at locations, and syncing our footage was a true exercise in project management and collaboration. Editing the animations of silk proteins and DNA insertion also gave me an appreciation for how visual storytelling can make an abstract molecular process feel tangible. Overall this assignment strengthened my media skills and reinforced how much planning goes into producing something that looks simple on screen.

## 5. Photos




# Assignment Three: Career Trajectory

## 1. Assignment Description

The assignment required me to investigate a person's career in biotechnology and explain their career progression and relation to their field. I have chosen to present the career of Frances Arnold, the 2018 Nobel Prize laureate in chemistry, who is known for her accomplishments in directed evolution and is also the first-ever chemical engineer to win the prestigious Nobel Prize in Chemistry. The presentation included her academic achievements, the issues she tackled in the scientific field, how her method of directed evolution works, the usefulness of her invention, as well as the companies and projects that were established due to her progress in research.

## 2. Summary of my Part Speech

Frances Arnold was born in 1956 in the city of Pittsburgh in Pennsylvania. She received a Bachelor of Science in Mechanical and Aerospace Engineering from Princeton University in 1979 and then a Ph.D. in Chemical Engineering from UC Berkeley in 1985, where she got excited about the newly emerging field of biotechnology. Arnold was later appointed the Linus Pauling Professor of Chemical Engineering, Bioengineering, and Biochemistry at Caltech and became one of only a few scientists to be elected to the National Academy of Sciences, Engineering, and Medicine. Arnold developed what is known as directed evolution, whereby random mutations are introduced into a gene, and through screening different versions of a gene, the best-performing genes are selected, thus allowing the evolution "to design better enzymes." Early naysayers scoffed at this idea, but this feat would ultimately earn her the Nobel Prize in Chemistry in 2018 for "the directed evolution of enzymes." Subsequently, she co-founded three companies based on her work: Gevo (renewable fuels and chemicals), Provivi (non-toxic pest control), and Aralez Bio (unnatural amino acids), and opted to make her techniques public rather than patent each of them, to push the progress of the entire field rather than just those of her own lab.

## 4. Reflection

Studying Frances Arnold's career changed my perspective on the topic of scientific problem-solving. What impressed me the most was the way that Arnold transformed her design challenge predicting what is seen as the most suitable mutations of an enzyme into a search problem and allowed nature's evolutionary processes to accomplish the work instead of attempting to outsmart it. Such a creative approach is something I did not associate with engineering before this assignment. It is also worth mentioning her persistence because despite the fact that her idea was dismissed as not scientific, Frances Arnold still pursued it for years till its success, which is underscored by the fact that she won a Nobel Prize. This has shown me that the credibility of scientific research is sometimes behind true advancement and that having faith in a rational proposition matters even if it is not accepted right away. By giving up the right to patent her ideas and openly publishing them, she influenced my perception of a research career. I understood that the value of research may not only be assessed by personal or commercial success, but also by the degree to which the finding is used by others. Studying her career made me understand better what kind of researcher I want to become: someone who appreciates not only practical knowledge and experience, but also perseverance and creativity in tackling problems.

## 3. Photos of my speaking in the presentation



# Assignment Four: Acceptance and Ethical Issues

## 1. Assignment Description

The task required members of our class to engage in a formal debate about the societal acceptability and ethical implications of different biotechnologies. The topic that was assigned to us was de-extinction, which involves the application of cloning techniques and genetic engineering; we argued against the practice. This experience involved examining the ecological consequences of de-extinction as well as its scientific feasibility from an ethical and regulatory standpoint.

## 2. Summary of my Part Spoken

During my debate segment, I examined the ecological ramifications and ethical implications surrounding the reintroduction of extinct species into modern ecosystems and was able to show that ecosystems have developed in their absence and that any resurrected species may appear as invasive, disturb the existing predator and prey relationships, or be incapable of adapting to the new habitat. I also mentioned the ethical dilemma behind the immense amount of money that is spent on escaping extinction while failing to save other extant endangered species. Furthermore, I touched on the animal welfare issue since it is questionable whether cloned and hybrid animals would be fit as a result of genetic modification and if it is ethical to create new animals for research and public attraction purposes. In conclusion, I stated that there are insufficient regulatory systems for the application of de-extinction technologies due to the possible ecological consequences of the use of this biotechnological application.

## 4. Reflection

Being the leader of the negative team compelled me to make a case that was able to withstand criticism instead of merely presenting a subjective opinion. The most effective move made by the affirmative team consisted of presenting their position as “conditional” rather than unlimited de-extinction. Thus, I had to refine my arguments in the counter-round and provide evidence that regulation is not going to eliminate welfare and environmental uncertainty. Instead, regulation is just additional control of the existing problem. Moreover, coming up with rebuttals in real time allowed me to enhance my ability to think quickly and remain calm even in uncomfortable situations. Although the victory was enjoyable, it is more important to emphasize the positive effect the exercise had on my skills of critical evaluation — comparing the scientific possibilities with ethical responsibilities. I also realized that I had to present not only the risks identified but also explain why the risks are more significant than the gains, as well as skill in the acceptance of the perspective of the affirmative side, which proves that even though I do not think that full de-extinction is justified, biotechnology research in the sphere of conservation is useful.

### 3. Photos of my Speaking in the Debate





## Conclusions

The present module provided me with an enhanced and pragmatic perception of biotechnology compared to my previous knowledge. In particular, I acquired knowledge about industrial (white) biotechnology that refers to the process of making use of enzymes and microorganisms in manufacturing so as to replace petroleum-based production in sectors such as textiles, food production, and waste treatment. It was interesting to explore the area of recombinant DNA technology that helps to insert genes responsible for producing useful proteins like spider silk into bacteria. The examination of Frances Arnold's work appeared to be another essential part of my studies as it illustrated how directed evolution helped to combine engineering methods and biological processes and apply them to solving issues traditional methods cannot solve. Finally, I worked on four different assignments aimed at

developing an understanding of biotechnology as a combination of many different techniques utilized in accordance with the objective. The de-extinction controversy required me to test my ethics. I needed to assess the relationship between animal welfare, ecological inconclusiveness, and other priorities of nature conservation together with the scientific possibility of de-extinction instead of considering “Can we do it?” and “Should we do it?” as equivalents. In addition to the knowledge of the science itself, this course gave me a chance to develop the communication skills necessary for the presentation of technical content, as well as working in teams, assessing data critically, and asserting my point of view.

### Portfolio Arrangement Strategy

The way this portfolio is organized conforms to the structure established in the course template, which gives markers a simple way of finding and evaluating all the necessary elements in this portfolio. The first page is composed of the Abstract and Acknowledgements, followed by the Table of Contents and the Objectives section, where the goals of each of the four assignments are presented in proper order. The assignments are outlined one by one, which gives all the assignments their own section and the same structure: introduction to the assignment, presentation handout, video handout or debate script, summary prepared when necessary, my own thoughts about each of the assignments, and evidence in the form of photographs. The four assignments are organized according to their completion time: the group presentation about white biotechnology, the video project about spider silk, the career presentation about Frances Arnold, and the ethics debate about de-extinction. The Conclusion is written after the last assignment in order to connect the outcomes reached throughout the whole course.

## Assessment Evaluation Sheet

Official Module Evaluation Criteria & Marking Rubric:

| Criteria | Marks Allocated | Marks Obtained |
|---|---|---|
| 1. Objectives | 10 | |
| 2. Depth and breadth (include all assignments) | 5 | |
| 3. Organized and Structured | 10 | |
| 4. Flow of information | 5 | |
| 5. Reflection | 15 | |
| 6. Conclusion/summary | 5 | |
| 7. Professional appearance | 10 | |
| 8. Creativity – beyond the template given | 30 | |
| 9. Visual appeal | 10 | |
| **Total Marks** | **100** | |